

equine Integration

Horse Training Feedback Generator

Feedback Document

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Iteration 1: AI Data Event at Fontys

Feedback Received:

- ***Graph Initialization:*** *The first 3–4 weeks of training data should not be included when calculating averages, as they do not reflect a stable training baseline.*
- ***Zero Values in Moving Averages:*** *A moving average should not drop to zero. If it does, this typically signals missing data or an anomaly.*
- ***Rest Days Interpretation:*** *Days with zero training load typically indicate a recovery period, not a lack of data.*
- ***Training Peaks:*** *Spikes in training load often occur when horses return to work after a period of rest or lighter activity. These are sometimes linked to owners unintentionally overtraining horses after a break.*

Actions Taken:

- *Adjusted the data preprocessing steps to exclude the first few weeks from baseline calculations.*
- *Updated logic to correctly interpret recovery days based on training load.*
- *Integrated detection for training spikes and overtraining risks after periods of inactivity.*

Iteration 2: First Product Review

Feedback Received:

Textual Feedback:

- ***Model Advice Issues:*** *The model recommended rest despite evidence of undertraining, which contradicted professional expectations.*
- ***Contradictory Suggestions:*** *Horses identified as undertrained were incorrectly advised to rest rather than gradually increasing workload.*

Visual Feedback:

- ***Lack of Visuals in Feedback:*** *Graphs should be included with each feedback message to clarify recommendations.*
- ***Visual Complexity:*** *Graphs were too technical and needed to be more user-friendly.*
- ***Zone Display:*** *The use of "zone" labels in the raw data was a good move for model for easier interpretation.*

- ***Language Accessibility:*** *Dutch translations for rider feedback were missing.*

Actions Taken:

- *Investigated and corrected the feedback generation logic to better reflect undertraining and recovery conditions.*
- *Personalize the feedback for each horse to match message style of writing and language (English/ Dutch).*
- *Performing comparative evaluations of multiple AI models to ensure higher accuracy and reliability.*
- *Merging additional datasets to capture more complete horse training profiles. (Optional based on research)*
- *Reworked the feedback visualization system to include:*
    - *A structured layout of text followed by graph explanations.*
    - *Simplified chart visuals by removing gridlines and reducing technical clutter.*
    - *Visual icons and short textual summaries beneath each chart to make interpretations easier.*